Ankit

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Education

National Institute of Technology, Raurkela **Aug 2025 – Present**
M.Sc. in Computational Mathematics and Data Science *CGPA: 7.80*

Relevant Coursework

- Data Structures and Algorithms
- Linear Algebra
- Optimization
- Probability and Statistics
- Mathematical Finance
- Numerical Analysis

Technical Skills

Programming & Querying: Python, SQL, C++
Data Preparation & Analytics: Pandas, NumPy, EDA, Data Cleaning, Data Validation, Feature Engineering, Statistical Analysis
Machine Learning & Statistics: Scikit-learn, Random Forest, XGBoost, Classification, Regression, Clustering, Model Evaluation, Cross-Validation
Deep Learning & Generative AI: PyTorch, TensorFlow, Keras, NLP, Prompt Engineering, LLM APIs, Ollama
Visualization & Applications: Power BI, Matplotlib, Seaborn, Plotly, Streamlit
Tools & Deployment: Git, GitHub, MySQL, FastAPI, Docker, Jupyter, Google Colab

Experience

Research Intern, BITS, Pilani **May 2026 – July 2026**
DNA Sequence Prediction using Deep Learning

- Built a **deep learning pipeline** to predict the next 3 nucleotides in the human **GRCh38 genome** from preceding sequence context, framing genomic prediction as an autoregressive language-modeling task.
- Designed and evaluated **CNN, BiLSTM, and CNN+BiGRU** architectures for next-nucleotide prediction, comparing per-base prediction accuracy against a 25% random baseline.
- Implemented a chromosome-level train/validation/test split to **prevent data leakage** and performed error analysis across repetitive and unique genomic regions using k-mer overlap and per-position accuracy.

Projects

Smart Cargo Monitoring and Decision Support System | GitHub **Sept 2026**
Python, Pandas, Scikit-learn, XGBoost, Streamlit, Ollama

- Performed **exploratory data analysis** on SCMS shipment records to identify late-delivery patterns, high-risk vendors, shipment-mode inefficiencies, and cost-related logistics issues, translating findings into operational insights.
- Built **Random Forest and XGBoost** delay-risk models with cost-per-kg, scheduled month, weekday, and missingness features; applied **time-based splitting, time-series cross-validation, class weighting, calibration, and threshold optimization**.
- Achieved **81.3% average ROC-AUC** and **73.7% accuracy** across three chronological back-testing periods; developed an interactive **Streamlit decision-support application** with calibrated risk probabilities, risk bands, model-driver explanations, and Ollama-powered operational recommendations.

Heatwave Prediction in Rajasthan | GitHub **March 2026**
Python, Scikit-learn, FastAPI, Docker, OpenWeather API

- Engineered an end-to-end **machine learning application** to predict heatwave conditions and assess outdoor work safety using **Random Forest** with Low/Medium/High risk classification.
- Developed **5 RESTful FastAPI endpoints** with Pydantic validation for real-time predictions, weather retrieval, and risk assessment, incorporating temperature-humidity interactions and heat index features.
- Containerized the application using **Docker** and integrated OpenWeather API data with a frontend interface, covering the full **development-to-deployment workflow** in a reproducible setup.

Sleep Breathing Analysis | GitHub **Jan 2026**
Python, TensorFlow, Keras, SciPy, Signal Processing

- Built an end-to-end **deep learning pipeline** to classify sleep breathing events including Hypopnea and Obstructive Apnea from PSG signals across 5 participants.
- Applied a **4th-order Butterworth bandpass filter** (0.17–0.4 Hz), segmented signals into 30-second windows with 50% overlap, and trained a **1D CNN** using Leave-One-Participant-Out Cross-Validation.
- Generated automated **PDF diagnostic reports** containing annotated PSG signal overlays and confusion matrices for participant-level evaluation.